Day 1 Notes: Time Complexity & Machine Learning Basics

Date: October 26, 2026 **Topics Covered:** DSA (Time Complexity), Machine Learning (Introduction), Statistics (Introduction).

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1. DSA: Time Complexity

Why do we learn Data Structures and Algorithms (DSA)? We receive data from the user side, and we need a meaningful way to arrange and organize that data. That is exactly why we need Data Structures and Algorithms.

What structures do we use to store data? We have **Arrays, Linked Lists, Stacks, and Queues**. These are all *linear* data structures. There are also *non-linear* data structures, such as **Trees and Graphs**.

- [![[Pasted image 20261026210534.png]] A simple diagram showing Linear vs Non-Linear data structures*

Then what exactly are Algorithms? Algorithms are structured sets of instructions that humans give to computers. These are the programs we use to solve a problem.

How do we decide which algorithm is faster and which one serves our use case? This is where **Time Complexity** comes in. It is the metric we use to measure the speed of an algorithm and understand how much time it takes to execute a task.

For example: If Algorithm A takes 10 seconds and Algorithm B takes 15 seconds, and we are building something that requires fast data processing, we will choose Algorithm A.

Understanding Time Complexity

Imagine you write 10 lines of code. One small part of that code takes 1 second to execute, but the other 90 lines take 10 seconds. What do you say the whole code takes? You say 10 seconds.

Why? Because that 1 second is *negligible* compared to the 10 seconds. We only care about the part of the code that scales and takes the most time.

The Order of Growth (Big O)

To compare algorithms, we need to know the order of growth. From fastest to slowest, the hierarchy looks like this:

$1 < \log(\log n) < \log(n) < n < n^2 < n^3 < n^4 < \dots < 2^n$

- **N** = the size of the input.
- If N is very, very large, the algorithm on the right side (like 2^n) will take an impossibly long time. The algorithm on the left side (like 1 or log n) will barely break a sweat.

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2. Machine Learning: Introduction (CampusX)

- 1 AI vs NN vs ML vs DL*

What is difference between each of these term this diagram below will show the difference : * [![[Pasted image 20261026210947.png]] Ai vs DL vs NN vs ML*

what is Ai ?

it can do its task without Any human intervanction this is the simeple and most genral defination of Ai

where the use of Ai and And Ml Algos mostely used ?

the Ai and ML Algos are mostely used in Social media Application to personilize our feeds and recommendation systems

Netflix --> Action/comedy movie --> Recommend like wise movies or series this Algos are high;y optimosed for that only.

what is ml

it is a subset of Ai to Analyze data , visualize data, forecast the data.

Deep learning

deep learning is simply Ml Algorithim used to mimic human Brain using Nurons

Data Science

Data Science is using the all skills Above mentioned to produce a meaning output from data is called Data Science

Types of machine Learning

Machine Learning Classification Table

Category	What it means	Common Algorithms	Real-world Use Case
Supervised Learning	The model learns from **labeled data**. (We give it the input *and* the correct answer).	• Linear Regression • Logistic Regression • Decision Trees • Random Forest • Support Vector Machines (SVM) • K-Nearest Neighbors (KNN)	Predicting house prices, spam email detection, image classification.
Unsupervised Learning	The model learns from **unlabeled data**. (We give it the input, but *no* answers; it finds patterns on its own).	• K-Means Clustering • Hierarchical Clustering • DBSCAN • Principal Component Analysis (PCA) • Apriori Algorithm	Customer segmentation, anomaly detection, reducing data dimensions.

Quick definitions to remember for your exams:

- **Supervised = "Teaching with an answer key." (Task: Classification & Regression)**

- **Unsupervised** = "Here is a pile of data, find the hidden groups." (Task: Clustering & Association)
- **Reinforcement (Forward)** = "Reward and Punishment." (Task: Decision Making)

2. Machine Learning: Core Concepts

Machine Learning is broadly divided into two main types based on how the algorithm learns: **Supervised Learning** and **Unsupervised Learning**.

A. Supervised Learning

Definition: The model is trained on a **labeled dataset**. This means every input data point comes with the correct answer (the "label" or "target"). The algorithm learns the relationship between the input and the output so it can predict the answer for new, unseen data.

How it works: You show the computer thousands of pictures of cats labeled "Cat" and dogs labeled "Dog." Later, when you show it a new picture, it uses the rules it learned to guess if it's a cat or a dog.

Two Main Types of Supervised Learning:

1. Regression

- **Definition:** Used when the output we are trying to predict is a **continuous number** (like price, temperature, or age).
- **Example:** Predicting the price of a house based on its square footage, number of bedrooms, and location.
- **Algorithms:** Linear Regression, Ridge/Lasso Regression.
- **Exam Key:** If the answer is a number on a sliding scale, it's regression.

2. Classification

- **Definition:** Used when the output we are trying to predict is a **discrete category** (like Yes/No, Spam/Not Spam, or Red/Green/Blue).
- **Example:** An email filter deciding if an incoming email is "Spam" or "Not Spam."
- **Algorithms:** Logistic Regression (despite the name, it's for classification), Decision Trees, Random Forest, Support Vector Machines (SVM).
- **Exam Key:** If the answer is a specific label or category, it's classification.

B. Unsupervised Learning

Definition: The model is trained on an **unlabeled dataset**. There are no "correct answers" provided. The algorithm's job is to discover hidden patterns, structures, or relationships in the data on its own.

How it works: You give the computer a massive pile of customer data (age, location, purchase history) without telling it anything about the customers. The computer groups them based on similarities.

Main Types of Unsupervised Learning:

1. Clustering

- **Definition:** Grouping data points together based on similarity. Data points in the same group (cluster) are more similar to each other than to those in other groups.
- **Example:** Customer segmentation. A company groups its customers into "Bargain Hunters," "Loyalists," and "Big Spenders" to target ads better.
- **Algorithms:** K-Means Clustering, Hierarchical Clustering, DBSCAN.

2. Dimensionality Reduction

- **Definition:** Reducing the number of features (variables) in a dataset while retaining as much information as possible. It makes the data easier to visualize and speeds up computation.
- **Example:** Compressing a dataset with 100 columns down to 3 principal components to plot it on a 3D graph.
- **Algorithms:** Principal Component Analysis (PCA), t-SNE.

3. Association

- **Definition:** Finding rules that describe how large portions of data are connected.
- **Example:** The classic "People who bought X also bought Y" (like Amazon's recommendation engine).
- **Algorithms:** Apriori, Eclat.

Summary: Supervised vs. Unsupervised

Feature	Supervised Learning	Unsupervised Learning
Data	Labeled (Input + Output)	Unlabeled (Only Input)
Goal	Predict the outcome	Discover hidden patterns

Feature	Supervised Learning	Unsupervised Learning
Tasks	Regression & Classification	Clustering & Dimensionality Reduction
Example	Predicting house prices	Grouping customers by buying habits

3. Statistics: Introduction

(Just started with basic starts not much to share still You can Refer this : statistics' In 1 hour **(basic) [English])**

References & Resources

1. **Time Complexity & Big O Notation [Hindi]**
    - **Chai Aur Code DSA**
2. **Machine Learning Basics [Eng]**
    - **100 Days of Machine Learning | CampusX**
3. **Statistics**
    - **Krish Naiks Stats in 6 Hour [Hindi]**

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